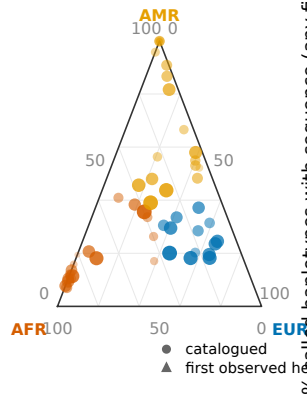


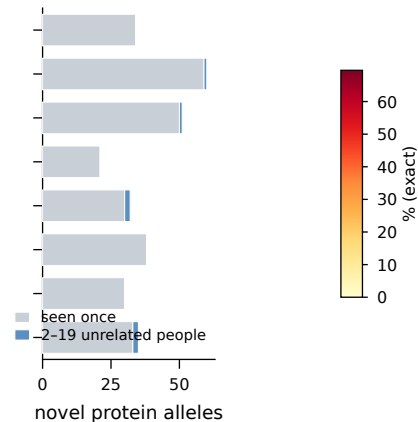
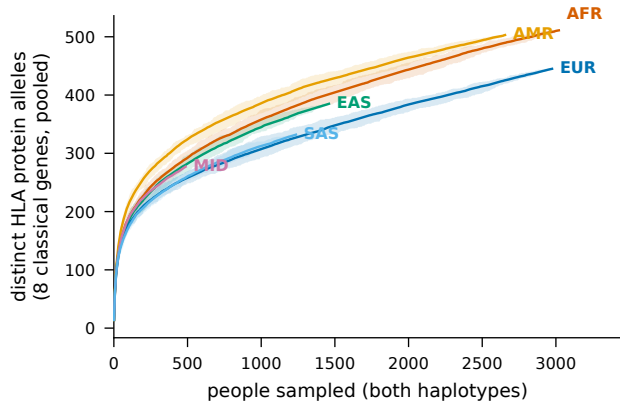
a

panel a: awaiting 40a_admixture_bins.py VM run

b**c**

% allel haplotypes with sequence (any field) absent from IPD-IMGT/HLA -- mostly non-coding

	AFR	AMR	EAS	EUR	MID	SAS
DRB1	≤66	≤67	≤72	≤46	≤44	≤67
DQB1	≤12	≤9	≤11	≤8	≤12	≤10
DQA1	≤17	≤10	≤18	≤13	≤21	≤21
DPB1	≤26	≤15	≤29	≤13	≤23	≤23
DPA1	≤13	≤13	≤15	≤9	≤17	≤15
C	≤9	≤7	≤9	≤7	≤10	≤10
B	12	≤11	≤10	≤8	≤13	≤12
A	≤4	≤3	≤4	≤5	≤7	≤4

d**e****f**